

L5 - Operational Security

Safe multisig 2-of-3 (demo) / 3-of-5 (prod)

Hardware wallet for governance signers

Key rotation policy (demo / dev / emergency)

L4 - Runtime Monitoring

Tenderly alerts (6 triggers)

OpenZeppelin Defender automation

The Graph subgraph + GraphQL

Structured logs + correlation IDs

L3 - AI / ML Defenses

RandomForest fraud + IsolationForest anomaly

SHAP explanations (top-3 factors stored)

Behavioural biometrics (session_events)

Federated learning across Local Banks

L2 - Application Security

EIP-712 wallet auth + 15-min JWT

Rate limiting (slowapi 5-60-10 / min)

CORS allowlist + CSP headers

Pydantic input validation

JWT blacklist in Redis

L1 - Smart Contract Security

UUPS proxy + TimeLock (24-48h)

AccessControl roles + role expiry

Reentrancy guards, granular pause

KYC + zkAML on-chain gates

Slither + Mythril + Echidna + Halmos / Certora